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Selected Publications & Manuscripts

ACCEPTED

1. PilotBench: A Benchmark for General Aviation Agents

Yalun Wu*, Haotian Liu*, Zhoujun Li, Boyang Wang

IEEE International Joint Conference on Neural Networks (IJCNN 2026). *arXiv:2604.08987*

A flight-agent benchmark built from 708 real-world trajectories and 34-channel telemetry. Evaluation across 41 models identifies a precision–controllability trade-off and a dynamic-complexity gap in high-workload flight phases. Pilot-Score combines regression accuracy with instruction following and constraint compliance. Traditional forecasters reach MAE 7.01, while LLMs achieve 86–89% instruction following at 11–14 MAE.

2. FLY-EVAL++: An Evidence-Driven Evaluation Protocol for Flight Prediction with Large Language Models

Yalun Wu, Junfeng Fang, Jiawei Wang, Haotian Liu, Qijun Yang, Minghan Yang, Hongcheng Guo, Zhoujun Li, Boyang Wang

Conference on Language Modeling (CoLM 2026). *arXiv:2609.04021*

An evidence-driven protocol for structured validity, physical feasibility, and constraint compliance in flight prediction. Across 66 LLMs, predictive quality and constraint compliance emerge as distinct capability dimensions, with a 28.2-point constraint gap at comparable predictive quality.

MANUSCRIPTS

3. Calibrated Lexical Fusion for Noise-Robust Retrieval of Colloquial Chinese Procurement Queries

Haotian Liu, Xiaowen Liu, Yunhan Wang, Yalun Wu, Minghan Yang, Xinhao Yu, Boyang Wang

ICASSP 2027 – Under review. First author.

A context-aware procurement-search system combining four-path hybrid retrieval, hotel-profile-conditioned ranking, and online Thompson Sampling. On 175 queries over a 1,854-SKU benchmark, it reaches MRR 0.95 and NDCG@5 0.91; the centralization predictor reaches MAE 0.064 and R-squared 0.747, while live A/B tests report 6.2% and 9.8% first-hit CTR gains.

4. Before the Agent Acts: Understanding, Task Accounts, and Implementation in Pre-Coding Review

Haotian Liu et al.

Manuscript in preparation.

Introduces Collaborative Evidence-Constrained Task Interpretation (CETI), a repository-grounded pre-coding review workflow with editable, source-linked task Records. A 24-participant study over 72 cases reports a +17.2 percentage-point adjusted Records-2 versus FreeChat-2 contrast in objective understanding; a linked audit traces 102 request occurrences through 93 families and 504 coding attempts across 10 tasks and 3 models. Exact criteria cover 20 of 24 core families and 4 of 25 boundary families.

* Equal contribution.